

3.0 Introduction

Relationships between elements of sets occur in many contexts. Relations can be used to solve problems involving communications networks, project scheduling, etc.

3.1 Boolean Matrix

An $m \times n$ matrix whose entries are either one or zero, is called a **Boolean Matrix**.

For example

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

is a Boolean matrix of order 3×4 .

3.1.1 Join and Meet of Boolean Matrices

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ be two Boolean matrices of same size.

The *join* of A and B, denoted by $A \vee B$, is a matrix $C = [c_{ij}]_{m \times n}$ where

$$c_{ij} = \begin{cases} 1, & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0, & \text{if } a_{ij} = 0 \text{ and } b_{ij} = 0 \end{cases}$$

Symbolically, we write

$$A \vee B = C.$$

The *meet* of A and B, denoted by $A \wedge B$, is a matrix $D = [d_{ij}]_{m \times n}$ where

$$d_{ij} = \begin{cases} 1, & \text{if } a_{ij} = 1 \text{ and } b_{ij} = 1 \\ 0, & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases}$$

Symbolically, we write

$$A \wedge B = D.$$

Remark 1. The join and meet of two Boolean matrices are also Boolean matrices.

Example 1. Compute the join and meet of the matrices:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Solution: Join of A and B ($A \vee B$)

Let $C = [c_{ij}]$ be the join of A and B. Since a_{31} and b_{31} , are both zero so by definition of join, $c_{31} = 0$ and

similarly, $a_{32} = 0 = b_{32}$ yield to $c_{32} = 0$. In remaining cases, either $a_{ij} = 1$ or $b_{ij} = 1$, so $c_{ij} = 1$.

Hence

$$C = A \vee B$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Meet of A and B ($A \wedge B$)

Let $D = [d_{ij}]$ be the meet of A and B. Since $a_{11} = 1 = b_{11}$; $a_{23} = 1 = b_{23}$; $a_{42} = 1 = b_{42}$ so by definition of meet, $d_{11} = 1$; $d_{23} = 1$; $d_{42} = 1$. In remaining cases, either $a_{ij} = 0$ or $b_{ij} = 0$, so corresponding $d_{ij} = 0$.

Hence

$$D = A \wedge B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

3.1.2 Product of Boolean Matrices

The *Boolean product* of two Boolean matrices $A = [a_{ij}]_{m \times p}$ and $B = [b_{ij}]_{p \times n}$ denoted by $A \odot B$, is the Boolean matrix $C = [c_{ij}]_{m \times n}$, where

$$c_{ij} = \begin{cases} 1, & \text{if } a_{ik} = 1 = b_{kj} \text{ for some } k, 1 \leq k \leq p \\ 0, & \text{if } a_{ik} = 0 \text{ or } b_{kj} = 0, \text{ for some } k, 1 \leq k \leq p \end{cases}$$

Symbolically, we write

$$A \odot B = C$$

Computation Procedure

Step 1. Select i^{th} row of A and j^{th} column of B.

Step 2. Compare the first element of the i^{th} row with the first element of the j^{th} column, second element of the row with the second element of the column and so on. If in any case, both elements are equal to 1, then $c_{ij} = 1$ otherwise $c_{ij} = 0$.

Example 2. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{4 \times 3}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}_{3 \times 4}$. Compute $A \odot B$.

Solution: Let

$$A \odot B = C \\ = [c_{ij}]_{4 \times 4}$$

$$= \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix}_{4 \times 4}$$

Now, on comparing first row of A with the first column of B, we observe that

Similarly,

$$\begin{aligned} a_{11} = 1 = b_{11} &\Rightarrow c_{11} = 1 \\ a_{12} = 1 = b_{22} &\Rightarrow c_{12} = 1; a_{12} = 1 = b_{23} \Rightarrow c_{13} = 1 \\ a_{22} = 1 = b_{22} &\Rightarrow c_{22} = 1; a_{22} = 1 = b_{23} \Rightarrow c_{23} = 1 \\ a_{31} = 1 = b_{11} &\Rightarrow c_{31} = 1; a_{32} = 1 = b_{22} \Rightarrow c_{32} = 1 \\ a_{32} = 1 = b_{23} &\Rightarrow c_{33} = 1; a_{43} = 1 = b_{31} \Rightarrow c_{41} = 1 \\ a_{43} = 1 = b_{33} &\Rightarrow c_{43} = 1; a_{43} = 1 = b_{34} \Rightarrow c_{44} = 1 \end{aligned}$$

In rest of the cases, $c_{ij} = 0$

$$\therefore A \odot B = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

3.1.3 Properties of Boolean Matrices

Let A, B and C are Boolean matrices of compatible sizes, then

(i) Commutative Property

1. $A \vee B = B \vee A$
2. $A \wedge B = B \wedge A$

(ii) Associative Property

1. $A \vee (B \vee C) = (A \vee B) \vee C$
2. $A \wedge (B \wedge C) = (A \wedge B) \wedge C$
3. $A \odot (B \odot C) = (A \odot B) \odot C$

(iii) Distributive Property

1. $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$
2. $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$

3.2 Ordered Pairs

While listing the objects of a set we do not assign their position and order. But several times we need pair of two objects a and b in which the position of these objects is also very important.

Let a and b be any two objects, a is assigned as the first position and b is assigned as the second position, then (a, b) is called the ordered pair of a and b .

... inverse relation.

3.11 R-Relative Set of an Element x

Let R be a relation from A to B and $x \in A$, then R relative set of x , denoted by $R(x)$, is the set of those elements of B which are related to that particular element x i.e.,

$$R(x) = \{y \in B \mid (x, y) \in R\}.$$

Example 9. Let

$$A = \{1, 2, 3, 4\}, B = \{r, s, t\}$$

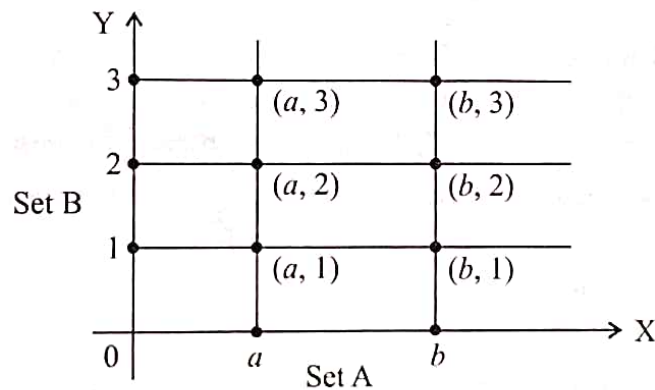
$$R = \{(1, s), (1, t), (2, 4), (2, s), (4, t)\}$$

$$R(1) = \{s, t\}; R(2) = \{r, s\}; R(4) = \{t\}.$$

3.12 Graph of a Relation

As a relation R from a set A to a set B is a subset of the product set $A \times B$, therefore R can be sketched on a co-ordinate diagram similar to that of the set $A \times B$.

Example 10. Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$, then $R = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$ is a relation from A to B . Its graph can be drawn as shown in the following figure.



3.13 Adjacency Matrix of a Relation

Let elements of sets A and B be labelled as

$$A = \{a_1, a_2, \dots, a_m\}$$

and

$$B = \{b_1, b_2, \dots, b_n\}.$$

Suppose R be a relation from A to B . Then the relation R can be represented by an $m \times n$ matrix

$$M_R = [m_{ij}]_{m \times n},$$

where

$$m_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R \\ 0, & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is known as the matrix of relation R or adjacency matrix or Boolean matrix of R .

Example 11. Let $X = \{1, 2, 3\}$, $Y = \{1, 2\}$ and $R = \{(2, 1), (3, 1), (3, 2)\}$ then M_R is a 3×2 matrix, whose elements m_{21} , m_{31} , m_{32} are each equal to unity and other elements are all zero,

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Example 12. Find the relation R , when

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

be a matrix of the relation R from X to Y , where $X = \{x_1, x_2, x_3\}$ and $Y = \{y_1, y_2, y_3, y_4, y_5\}$.

Solution. The relation is

$$R = \{(x_1, y_2), (x_2, y_1), (x_2, y_3), (x_2, y_4), (x_3, y_1), (x_3, y_3), (x_3, y_5)\}.$$

3.14 Representing Relations using Digraphs

If R be a relation on a finite set A then R can be represented pictorially. In this representation each element of the set is represented by a point and each ordered pair (a, b) is represented by an arc (or edge) directed from a to b . Such representations are called **digraphs** or **directed graphs**.

An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a loop.

Indegree of a vertex : If R be a relation on a set A and $v \in A$ then indegree of v is the number of edges directed towards v .

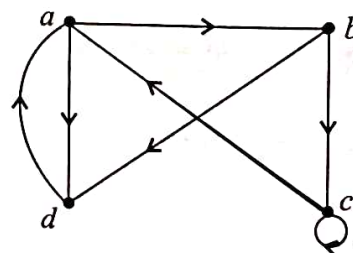
Outdegree of a vertex : If R be a relation on a set A and $v \in A$ then outdegree of v is the number of edges beginning from v i.e., going away from v .

Example 13. Let $A = \{a, b, c, d\}$ and

$$R = \{(a, b), (a, d), (b, d), (c, a), (c, b), (c, c), (d, a)\}$$

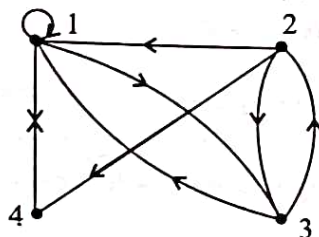
then digraph of R can be displayed as below

Here



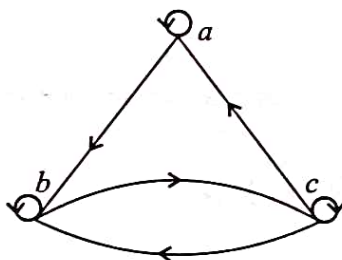
Vertices	:	a	b	c	d
Indegree	:	2	1	2	2
Outdegree	:	2	2	2	1

Example 14. Find the relation R , defined on the set $A = \{1, 2, 3, 4\}$, represented by the digraph



Here $R = \{(1, 1), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1)\}$

Example 15. Let R be a relation on the set $A = \{a, b, c\}$. Find R and matrix representation M_R of R , whose digraph is given below



Solution. Here $R = \{(a, a), (a, b), (a, c), (b, b), (b, c), (c, a), (c, b), (c, c)\}$

$$\therefore M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

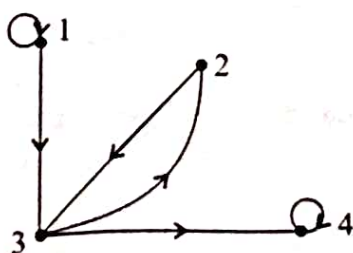
Example 16. If a relation R on the set $A = \{1, 2, 3, 4\}$ has the matrix representation

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Draw the digraph of R .

Solution. The relation $R = \{(1, 1), (1, 3), (2, 3), (3, 2), (3, 4), (4, 4)\}$

$\therefore$ Digraph is as follows



3.14

Vertices	:	1	2	3	4
Indegree	:	1	1	2	2
Outdegree	:	2	1	2	1

3.15 Binary Relation

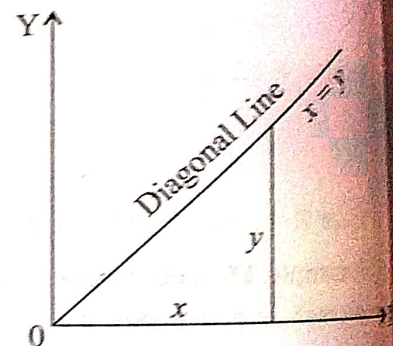
A relation R between pairs of elements of a set A is called **binary relation**.

In the discussions here after the word relation will mean binary relation.

Let A be any set and R be the set of all those pairs (x, y) of $A \times A$ for which $x = y$. Then the relation R is known as the **equality relation** in the set A . It is also known as the **diagonal relation** in A and is denoted by Δ .

Thus $\Delta = \{(x, y) \mid x = y, x, y \in A\}$.

Remark 3. Any subset of $A \times A$ is called a binary relation on the set A .

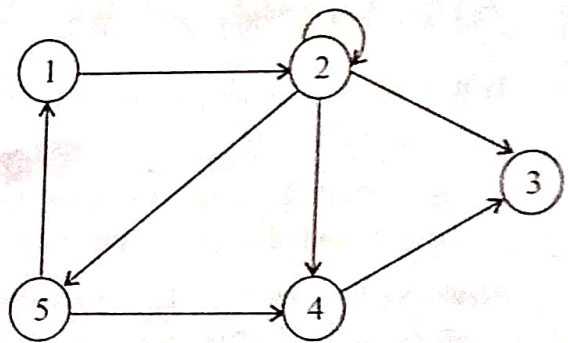


3.17 Connectivity Relation

Let R be a relation on a set A . A path of length n in R from a to b is a finite sequence $\Pi : a, x_1, x_2, \dots, x_{n-1}, b$, beginning with a and ending with b , such that

$$a R x_1, x_1 R x_2, \dots, x_{n-1} R b.$$

A path of length ' n ' involves $(n + 1)$ elements of the set A , although they are not necessarily distinct. For example, in the digraph



$\Pi_1 : 1, 2, 5, 4, 3$ is a path of length 4 from vertex 1 to vertex 3.

$\Pi_2 : 1, 2, 5, 1$ is a path of length 3 from vertex 1 to itself.

A path whose first and the last vertex are same is called a cycle. In above example, Π_2 is a cycle of length 3.

If n is a fixed positive integer, we can define a relation R^n on A as $a R^n b$ it implies that there is a path of length n from a to b in R .

We can also define a relation R^∞ on A as $a R^\infty b$ it implies that there is some path in R from a to b , whose length depends on a and b . The relation R^∞ is called the connectivity relation for R . If A has n elements then $R^\infty = R \cup R^2 \cup R^3 \cup \dots \cup R^n$.

Example 26. Given $A = \{1, 2, 3, 4\}$, consider following relation in A .

$$R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}.$$

Find (i) $R^2 = R \cdot R$.

(ii) R^∞

[Raj. 2003]

Solution. The digraph of R is as follows

Since $1 R 1$ and $1 R 1 \Rightarrow 1 R^2 1$

$$2 R 2 \text{ and } 2 R 2 \Rightarrow 2 R^2 2$$

$$3 R 2 \text{ and } 2 R 3 \Rightarrow 3 R^2 3$$

$$4 R 4 \text{ and } 4 R 2 \Rightarrow 4 R^2 2$$

$$4 R 4 \text{ and } 4 R 4 \Rightarrow 4 R^2 4$$

$$4 R 2 \text{ and } 2 R 3 \Rightarrow 4 R^2 3$$

$$2 R 2 \text{ and } 2 R 3 \Rightarrow 2 R^2 3$$

$$3 R 2 \text{ and } 2 R 2 \Rightarrow 3 R^2 2$$

$$\therefore R^2 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4), (4, 2), (4, 3)\}$$

(ii) Since $R^\infty = R \cup R^2 \cup R^3 \cup R^4$

Now,

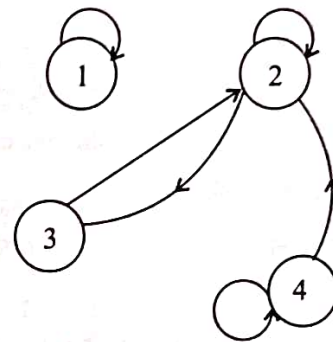
$$R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$$

$$R^2 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4), (4, 2), (4, 3)\}$$

$$R^3 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4), (4, 2), (4, 3)\}$$

$$R^4 = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4), (4, 2), (4, 3)\}$$

$$R^\infty = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4)\}$$



3.18 Closure of Relations

If R is relation defined on a set A then R may skip some important property P such as reflexivity, symmetry or transitivity.

A relation $R^\oplus$ is said to be **closure** of R with respect to property P , if $R^\oplus$ is the smallest relation that contains R and also satisfy the property P .

Reflexive Closure: Let R be a relation on A , then $R^\oplus$, reflexive closure of R , can be formed by adding to R all those pairs of the form (a, a) , $a \in A$, that are not present in R .

Symbolically, $R^\oplus = R \cup \Delta$, where Δ is the set of pairs of diagonal elements in the set A .

$$\Delta = \{(a, a) \mid \forall a \in A\}$$

Symmetric Closure: A relation $R^\oplus$, symmetric closure of R , can be formed by adding to R all those pairs of the form (b, a) that are absent from R while (a, b) present in R .

Symbolically, $R^\oplus = R \cup R^{-1}$, where R^{-1} is the inverse relation of R .

Transitive Closure: If R is a relation in a set A and if $|A| = n$, then transitive closure is the connectivity relation R^∞ .

Symbolically, $R^\oplus = R \cup R^2 \cup R^3 \cup \dots \cup R^n$.

Example 27. Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 2), (2, 3), (3, 4)\}$ be a relation in A . Find its reflexive closure, symmetric closure and transitive closure.

Solution: Reflexive Closure

If the set $A = \{1, 2, 3, 4\}$

then the set of diagonal pairs is

$$\Delta = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

So reflexive closure $R^\oplus$ is

$$R^\oplus = R \cup \Delta = \{(1, 2), (2, 3), (3, 4), (1, 1), (2, 2), (3, 3), (4, 4)\}$$

Symmetric Closure

Let

$$R = \{(1, 2), (2, 3), (3, 4)\}$$

then

$$R^{-1} = \{(2, 1), (3, 2), (4, 3)\}$$

So by symmetric closure, we have

$$R^\oplus = R \cup R^{-1} = \{(1, 2), (2, 3), (3, 4), (2, 1), (3, 2), (4, 3)\}$$

Transitive Closure

If

$$R = \{(1, 2), (2, 3), (3, 4)\}$$

and the order of set A i.e., $|A| = 4$, then we find

$$R^2 = \{(1, 3), (2, 4)\}$$

$$R^3 = \{(1, 4)\}$$

$$R^4 = \phi$$

so transitive closure $R^\oplus$ is given by

$$R^\oplus = R \cup R^2 \cup R^3 \cup R^4 = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}$$

3.19 Transitive Closure: Warshall's Algorithm

[RTU 2010]

In 1960, Stephen Warshall described an algorithm, known as Warshall's algorithm for computing the transitive closure of a relation.

Let R be a relation on a set A with n elements say $a_1, a_2, \dots, a_n$. If $a, x_1, x_2, \dots, x_{m-1}, b$ is a path in R then the vertices $x_1, x_2, \dots, x_{m-1}$ i.e., the vertices other than the first and the last vertex of the path, are called interior vertices of the path.

If in the above path, any $x_i = a$, $1 \leq i \leq m-1$, then a can be considered as an interior vertex. Similarly, if any $x_j = b$, $1 \leq j \leq m-1$, then b can also be considered as an interior vertex.

This algorithm is based on the construction of a sequence of Boolean matrices, say $W_0, W_1, \dots, W_m$ where $W_0 = M_R$ (matrix of a given relation R), $W_n = M_{R^n}$ (matrix of a desired transitive closure of R).

and

$$W_k = [w_{ij}^{(k)}], \text{ where}$$

$$w_{ij}^{(k)} = \begin{cases} 1, & \text{if there is a path from } a_i \text{ to } a_j \text{ in } R \text{ such that} \\ & \text{all the interior vertices of this path are in set} \\ & \{a_1, a_2, \dots, a_k\} \text{ if any} \\ 0, & \text{Otherwise} \end{cases}$$

Since every interior vertex belongs to the set $\{a_1, a_2, \dots, a_n\}$, it follows that the (i, j) th entry of W_n is 1 if and only if there is a path from a_i to a_j in R . Thus $W_n = M_{R^n}$.

Next, we will show that we can compute the matrix W_k from the previous matrix W_{k-1} . So that we can begin with the matrix $W_0 (=M_R)$ and proceed one step at a time until, in n steps, we obtain the matrix $W_n (=M_{R^n})$. This procedure is called *Warshall's algorithm*.

Let $W_k = [y_{ij}]$ and $W_{k-1} = [x_{ij}]$. Now if $y_{ij} = 1$, then there must be a path from a_i to a_j such that the interior vertices of this path are from the set $\{a_1, a_2, \dots, a_k\}$. If a_k is not an interior vertex [figure (a)] then the interior vertices of the path are from the set $\{a_1, a_2, \dots, a_{k-1}\}$ and then $x_{ij} = 1$. If a_k is an interior vertex then it appears exactly one time in the path, assuming that all interior vertices are distinct, and then all the interior vertices of the path from a_i to a_k and from a_k to a_j must come from set $\{a_1, a_2, \dots, a_{k-1}\}$ figure (b), i.e., $x_{ik} = 1$ and $x_{kj} = 1$.

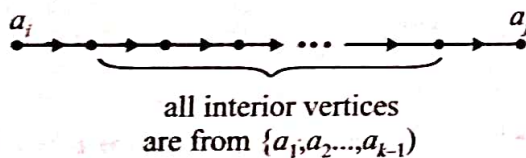


Fig. (a)

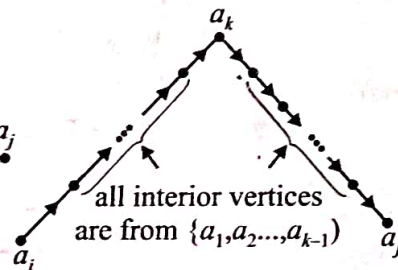


Fig. (b)

Thus, we can conclude that $y_{ij} = 1$ if and only if either

$$(1) x_{ij} = 1, \text{ or } (2) x_{ik} = 1 \text{ and } x_{kj} = 1.$$

So, if (i, j) th entry of W_{k-1} is 1 then by condition (1), (i, j) th entry of W_k is also 1. While from condition (2), if column k of W_{k-1} has a 1 in position i and row k of W_{k-1} has a 1 in position j , then (i, j) th entry of W_k must be 1.

Working Procedure to Compute W_k from W_{k-1}

Step 1. Copy all the 1's of W_{k-1} in W_k .

Step 2. Identify the position $s_1, s_2, \dots$, in column k of W_{k-1} , where the entry is 1, and the positions $t_1, t_2, \dots$, in row k of W_{k-1} , where the entry is 1.

Step 3. Write 1 at all the positions (s_p, t_j) of W_k , if 1 is not already written there.

Example 28. Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$ be a relation on A . Find the transitive closure of R using Warshall's algorithm.

Solution: Here

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = W_0$$

and

$$n = |A| = 4$$

Now, we find W_1 i.e., here $k = 1$. Column 1 of W_0 has a 1 at position 2 while row 1 of W_0 has a 1 at position 2. Thus W_1 has a 1 at (2, 2) position along with all the 1's of W_0 .

$$\therefore W_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Next, we find W_2 , i.e., for $k = 2$. Column 2 of W_1 has 1's at positions 1 and 2, while row 2 of W_1 has 1's at positions 1, 2 and 3. Thus W_2 has 1's at (1, 1), (1, 2), (1, 3), (2, 1), (2, 2) and (2, 3) positions, along with the existing 1's in W_1 .

$$\therefore W_2 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Further, we compute W_3 , i.e., for $k = 3$. Column 3 of W_2 has 1's at positions 1 and 2, while row 3 of W_2 has a 1 at position 4. So W_3 has 1's at (1, 4) and (2, 4) positions along with the existing 1's in W_2 .

$$\therefore W_3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

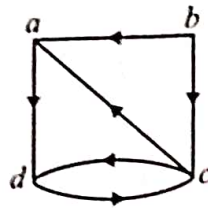
Finally, we compute W_4 , i.e., for $k = 4$. Column 4 of W_3 has 1's at positions 1, 2 and 3, while row 4 does not have 1 at any position. So there is no new 1's to be added in W_3 to obtain W_4 .

$$\therefore W_4 = W_3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M_{R^\infty}$$

Hence

$$R^\oplus = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)\}.$$

Example 29. Let R be the relation with digraph shown below. Find the transitive closure of R using Warshall's algorithm.



Solution: Here, set $A = \{a, b, c, d\}$ so $n = |A| = 4$.

Also,

$$R = \{(a, d), (b, a), (b, c), (c, a), (c, d), (d, c)\}$$

$$\therefore M_R = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = W_0$$

First, we compute W_1 , i.e., for $k = 1$. Column 1 of W_0 has 1's at 2 and 3 positions, while row 1 of W_0 has a 1 at position 4. Thus W_1 has 1's at (2, 4) and (3, 4) positions along with the existing 1's in W_0 .

$$\therefore W_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Next, we compute W_2 , i.e., for $k = 2$. Column 2 of W_1 has no 1's so there is no new 1 to be added to W_1 to obtain W_2 . Thus $W_2 = W_1$.

$$\therefore W_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Further, we compute W_3 , i.e., for $k = 3$. Column 3 of W_2 has 1's at positions 2 and 4, while row 3 of W_2 has 1's at positions 1 and 4. Thus W_3 has 1's at (2, 1), (2, 4), (4, 1) and (4, 4) positions along with the existing 1's in W_2 .

$$\therefore W_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Finally we compute W_4 , i.e., for $k = 4$. Column 4 of W_3 has 1's at positions 1, 2, 3 and 4, while row 4 of W_3 has 1's at positions 1, 3, and 4. Thus W_4 has 1's at (1, 1), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3) and (4, 4) positions along with the existing 1's in W_3 .

$$\therefore W_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} = M_{R^\infty}$$

$$\therefore R^\oplus = R^\infty = \{(a, a), (a, c), (a, d), (b, a), (b, c), (b, d), \\ (c, a), (c, c), (c, d), (d, a), (d, c), (d, d)\}$$

which is the desired transitive closure of R.

since $1 + 1 \wedge (-1) = 0$ (not +ve)

Hence R is reflexive and symmetric but not transitive.

- **Example 38.** Consider a set $A = \{a, b, c, d, e, f\}$ and a relation R defined on A given by $R = \{(a, a), (a, b), (b, a), (b, b), (c, c), (d, d), (d, e), (d, f), (e, d), (e, e), (e, f), (f, d), (f, e), (f, f)\}$

Write the matrix representation M_R of the relation and hence prove that it is an equivalence relation.

Solution: The set $A = \{a, b, c, d, e, f\}$.

The matrix representation of the given relation R is

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

- (i) **Reflexive :** The diagonal elements of the matrix M_R are non-zero it implies that every element of the set A is related to itself hence the relation is reflexive.
- (ii) **Symmetric :** The matrix M_R and its transpose $(M_R)^T$ are identical hence the relation is symmetric.
- (iii) **Transitive :** The matrix $(M_R)^2$ is given by

$$M_R \cdot M_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$= M_R$ (keeping in view $1 + 1 = 1$)

So we notice that $M_R M_R = M_R$.

Thus relation is transitive.

Hence the relation is an equivalence relation.

3.24 Equivalence Classes

If X be a set, R be an equivalence relation on X and $x \in X$, then the set of all those members $y \in X$, for which $x R y$, is called equivalence class of x and is denoted by

$$[x] = \{y \mid x, y \in X, x R y\} = [y]$$

$$y \in [x] \Leftrightarrow x R y$$

Thus, an equivalence class is a subset of the given set, such that any two elements of it are equivalent to each other.

Remark 10. Some authors write $\frac{x}{R}$ for the equivalence class x with respect to R . The set of all equivalence classes will be denoted by $\frac{X}{R}$ and read as “ X modulo R ” or “ X mod R ”. This is called the quotient set of X by R . Thus, the set of all disjoint equivalence classes defined by an equivalence relation R over a set S , is called the quotient set of S relative to R .

Example 43. Let X be the set of all triangles in a plane, and the equivalence relation R is for similarity. If $x, y \in X$, then by $[x]$ we mean the class or set of all those triangles which are similar to the triangle x and by $[y]$ we mean the class or set of all those triangle which are similar to the triangle y .

All the equivalence classes form a subset of $P(X)$, the assertion can be proved by noting that a subset A of $P(X)$ is an equivalence class if and only if for any $x \in A$, $[x] = A$, which is the quotient set of x modulo.

3.25 Properties of Equivalence Classes

Theorem 4 : Let A be a non-empty set and let R be an equivalence relation in A . Let x and y be arbitrary elements in A .

Then

- (i) $x \in [x]$
- (ii) If $y \in [x]$, then $[y] = [x]$,
- (iii) $[x] = [y] \Leftrightarrow (x, y) \in R$ i.e., iff $x R y$,
- (iv) Either $[x] = [y]$ or $[x] \cap [y] = \phi$ i.e., two equivalence classes are either disjoint or identical.

Proof.

- (i) Since R is reflexive, $x R x$, $\forall x \in x$

$$\text{But } [x] = \{a \mid a \in A \text{ and } a R x\}$$

$$\text{Thus, } x R x \Rightarrow x \in [x].$$

- (ii) $y \in [x] \Rightarrow y R x$, $\forall x, y \in A$

Let a be any arbitrary element of $[y]$, then $a \in [y] \Rightarrow a R y$.

But R is transitive, hence $a R y$ and $y R x$

$$\Rightarrow a R x \Rightarrow a \in [x]$$

$$\text{Thus, } a \in [y] \Rightarrow a \in [x]$$

$$\therefore [y] \subseteq [x]$$

.....(1)

Again let b be any arbitrary element of $[x]$, then

$$b \in [x] \Rightarrow b R x.$$

Since R is symmetric, $y R x \Rightarrow x R y$

$$\text{Now } b R x \text{ and } x R y \Rightarrow b R y$$

[$\because R$ is transitive]

$$\Rightarrow b \in [y]$$

$$\text{Thus, } b \in [x] \Rightarrow b \in [y]$$

$$\therefore [x] \subseteq [y]$$

.....(2)

$$\therefore (1) \text{ and } (2) \Rightarrow [x] = [y].$$

(iii) Let $[x] = [y]$, then we have to prove $x R y$ first.

As R is reflexive, $x R x$ and therefore

$$x R x \Rightarrow x \in [x]$$

$$\Rightarrow x \in [y]$$

$$[\because [x] = [y]]$$

$$\Rightarrow x R y$$

$$\text{Thus, } [x] = [y] \Rightarrow x R y.$$

Conversely : Suppose $x R y$, then we have to prove $[x] = [y]$.

Let a be any arbitrary element of $[x]$, then $a R x$ but it is given that $x R y$, therefore,

$$a R x \text{ and } x R y \Rightarrow a R y$$

$$[\because R \text{ is transitive}]$$

$$\Rightarrow a \in [y]$$

$$\text{Hence } a \in [x] \Rightarrow a \in [y] \text{ i.e., } [x] \subseteq [y]$$

.....(1)

Again let b be any arbitrary element of $[y]$, then

$$b \in [y] \Rightarrow b R y$$

$$\text{Since } R \text{ is symmetric, } x R y \Rightarrow y R x$$

$$\therefore b R y \text{ and } y R x \Rightarrow b R x$$

$$[\because R \text{ is transitive}]$$

$$\Rightarrow b \in [x]$$

$$\therefore b \in [y] \Rightarrow b \in [x] \text{ i.e., } [y] \subseteq [x]$$

.....(2)

$$\text{Hence } (1) \text{ and } (2) \Rightarrow [x] = [y]$$

$$\text{Finally } [x] = [y] \Rightarrow x R y \text{ and } x R y \Rightarrow [x] = [y]$$

$$\therefore [x] = [y] \text{ iff } x R y.$$

(iv) If $[x] \cap [y] = \phi$ then the result is obvious. Hence let us take $[x] \cap [y]$ non-empty and we have to prove $[x] = [y]$.

$$[x] \cap [y] \neq \phi \Rightarrow \exists x \in A \text{ s.t. } x \in [x] \cap [y].$$

$$\text{Now } a \in [x] \cap [y] \Rightarrow a \in [x] \text{ and } a \in [y]$$

$$\Rightarrow a R x \text{ and } a R y$$

$$\Rightarrow x R a \text{ and } a R y$$

$$[\text{as } R \text{ is symmetric, } a R x \Rightarrow x R a]$$

$$\Rightarrow x R y$$

$$[\because R \text{ is transitive}]$$

$$\Rightarrow [x] = [y]$$

[by (3)]

$$\text{Thus } [x] \cap [y] \neq \phi \Rightarrow [x] = [y]$$

$$\text{or } [x] \neq [y] \Rightarrow [x] \cap [y] = \phi.$$

• 3.26 Partition of a Set

Let X be a non-void set. A set $P = \{A, B, C, \dots\}$ of non-void subsets of X is called a partition of X if

1. $X = A \cup B \cup C \cup \dots$, i.e., the set X is the union of the sets in P , and
2. the intersection of every pair of distinct subsets of $X \in P$ is the null set i.e., if A and $B \in P$ then either $A = B$ or $A \cap B = \phi$.

Example 44. Consider the set $X = \{1, 2, 3, \dots, 9, 10\}$ and its subsets

$$B_1 = \{1, 3\}, B_2 = \{7, 8, 10\}, B_3 = \{2, 5, 6\}, B_4 = \{4, 9\}$$

The set $P = \{B_1, B_2, B_3, B_4\}$ is such that

- (i) B_1, B_2, B_3, B_4 are all non-void subsets of X ,
- (ii) $B_1 \cup B_2 \cup B_3 \cup B_4 = X$
- (iii) For any disjoint sets B_i, B_j , $B_i \cap B_j = \phi$

Hence the set $\{B_1, B_2, B_3, B_4\}$ is a partition of X .

Theorem 5 : An equivalence relation defined in set S decomposes the set into disjoint classes.

[RTU 2011, 2009]

Proof. Let an equivalence relation R be defined in a set S . Let $a \in S$ and T be a subset of S consisting of all those elements which are equivalent to a i.e.,

$$T = \{x \mid x \in S \text{ and } a R x\}.$$

Then $a \in T$, for $a R a$ (R is reflexive). Any two elements of T are equivalent to each other, for if $x, y \in T$, then $x R a$ and $y R a$.

$$\text{Again } x R a, y R a \Rightarrow x R a, a R y$$

[R is symmetric]

$$\Rightarrow x R y$$

Thus T is an equivalence class.

Let T_1 be another equivalence class i.e., $T_1 = \{x \mid x \in S \text{ and } x R b\}$,

where b is not equivalent to a . Then the classes T and T_1 must be disjoint. For if they have a common element s , $s R a$ and $s R b$, so that $b R a$ which is contrary to our hypotheses.

The set S can now be decomposed into equivalence classes $T, T_1, T_2, \dots$ such that every element of S belongs to one of these classes. Since these classes are mutually disjoint, we obtain the required partition of S .

Theorem 6 : An equivalence relation R in a non-empty set S determines a partition of S .

[IT(RTU-2009)]

Proof. Let R be an equivalence relation in S . Let A be the set of equivalence classes of S w.r.t. R i.e. $A = \{[a] : a \in S\}$ where $[a] = \{x : x \in S \text{ and } a R x\}$.

Now, R is an equivalence relation. Therefore $\forall a \in S$, we have $a R a$ (by reflexivity). Hence $a \in [a] \Rightarrow [a] \neq \phi$.

Further every element ' a ' of S is an element of the equivalence class $[a]$ in A . From this we conclude that $S = \bigcup_{a \in S} [a]$.

Finally, if $[a]$ and $[b]$ are two equivalence classes then either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Hence, A is a partition of S .

Example 45. Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the equivalence relation on A defined by

$$R = \{(1, 1), (1, 5), (2, 2), (2, 3), (2, 6), (3, 2), (3, 3), (3, 6), \\ (4, 4), (5, 1), (5, 5), (6, 2), (6, 3), (6, 6)\}$$

Find the partition of A induced by R i.e., find the equivalence classes.

[Raj. 2003]

Solution: The equivalence class $[1]$ is the set of elements of A related to 1 and these are 1 and 5, so

$$[1] = \{1, 5\}, \text{ similarly other classes are}$$

$$[2] = \{2, 3, 6\}$$

$$[3] = [2] = \{2, 3, 6\}$$

$$[4] = \{4\}$$

$$[5] = [1]$$

$$[6] = [2]$$

Hence the partition of A induced by R is

$$\{[1], [2], [4]\} = \{\{1, 5\}, \{2, 3, 6\}, \{4\}\}$$

Example 46. Consider $A = \{1, 2, 3, 4\}$ and let R be an equivalence relation

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4)\}. \text{ Find } \frac{A}{R}.$$

Solution. The equivalence classes are

$$[1] = \{1, 2\}$$

$$[3] = \{3, 4\}$$

So

$$\frac{A}{R} = \{\{1, 2\}, \{3, 4\}\}$$

Quotient set of A by R : The partition P of A obtained by an equivalence relation R on A is called quotient

set of A and this partition is represented by $\frac{A}{R} = \{[a] \mid a \in A\}$.

3.27 Product of Equivalence Relations

Let A and B be two sets and R, S be equivalence relations in A and B respectively. The relation $R \times S$ in $A \times B$, defined by $R(x, y) \times S(z, u) \Leftrightarrow x R u$ and $y S z$, is an equivalence relation in $A \times B$.

$R \times S$ is called the product of the relation R and S . Every equivalence class of $A \times B$ is of the form $P \times Q$, where P is an equivalence class of $X \bmod R$ and Q is an equivalence class of $Y \bmod S$.

3.28 Number of Partitions of a Finite Set

Suppose X be a finite set of size n . Then the number of partitions of X (or the number of equivalence relations on X) is given by $\sum_{r=1}^n S(n, r)$, where $S(n, r)$ denotes a stirling number of the second kind, defined as

$$S(n, 1) = 1 = S(n, n)$$

and

where $1 < r < n$.

$$S(n, r) = S(n-1, r-1) + r S(n-1, r),$$

Example 47. Compute the number of partitions of a set with

(i) Four elements

(ii) Five elements

Solution:

(i) Here $n = 4$

$$\text{Thus the required number of partitions} = \sum_{r=1}^4 S(4, r)$$

$$= S(4, 1) + S(4, 2) + S(4, 3) + S(4, 4) \quad \dots(1)$$

Now

$$S(4, 1) = 1 = S(4, 4) \quad \dots(2)$$

$$S(4, 2) = S(3, 1) + 2S(3, 2) \quad \dots(3)$$

$$S(4, 3) = S(3, 2) + 3S(3, 3) \quad \dots(4)$$

Since

$$S(3, 2) = S(2, 1) + 2S(2, 2) \quad \dots(5)$$

Now,

$$S(2, 1) = 1 = S(2, 2) \quad [\because S(n, 1) = 1 = S(n, n)]$$

Thus equation (5) gives

$$S(3, 2) = 1 + 2(1) = 3 \quad \dots(6)$$

$\therefore (3) \Rightarrow$

$$S(4, 2) = 1 + 2(3) = 7 \quad [\text{using (6) and } S(3, 1) = 1]$$

(4) $\Rightarrow$

$$S(4, 3) = 3 + 3(1) = 6$$

Hence the required number of partitions

$$= 1 + 7 + 6 + 1$$

$$= 15$$

(ii) Here $n = 5$

Thus the required number of partitions

$$= \sum_{r=1}^5 S(5, r)$$

$$= S(5, 1) + S(5, 2) + S(5, 3) + S(5, 4) + S(5, 5) \quad \dots(1)$$

Now,

$$S(5, 1) = 1 = S(5, 5)$$

Further,

$$S(5, 2) = S(4, 1) + 2S(4, 2)$$

$$= 1 + 2(7)$$

$$= 15$$

[from part (i)]

$$S(5, 3) = S(4, 2) + 3S(4, 3)$$

$$= 7 + 3(6)$$

$$= 25$$

[from part (i)]